Interpretable Multimodal Melanoma Classification — Project Walkthrough

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A step-by-step account of the methods, choices and results

Objective

The project rebuilds a first-year deep-learning exercise into a research-grade, **interpretable** classifier that distinguishes malignant from benign skin lesions on the public **ISIC 2020** dataset. The goal was not peak accuracy but a *methodologically honest* pipeline: correct validation, transparent handling of class imbalance, a fair test of whether clinical metadata help, and model interpretability — the qualities that separate a demonstration from a credible research artifact. All code is reproducible and public on GitHub.

Dataset. ISIC 2020: 33,126 dermoscopic images with a strong class imbalance (~1.8 % malignant) and per-patient clinical metadata (sex, approximate age, anatomical site).

Step-by-step pipeline

1. Patient-level data splitting (no leakage). A single patient contributes many lesions, so a naïve image-level split would place the same patient in both training and test sets and inflate performance. Splits were built with `StratifiedGroupKFold`, grouping on `patient_id` and stratifying on the label, so **no patient appears in more than one split**. This is the single most important methodological safeguard in the project.

2. Preprocessing and augmentation. Images were resized and normalised (ImageNet statistics). Training used light augmentation (horizontal/vertical flips, 90° rotations) via `albumentations`; validation and test used resize + normalise only. Crucially, all standardisation is fit on training data alone, applied to validation/test — no information leaks from held-out data.

3. Backbone and transfer learning. The image model is an **EfficientNet-B3** pretrained on ImageNet (via `timm`), with a lightweight classification head. Transfer learning is the appropriate choice given a few thousand positive examples: the backbone already encodes generic visual features, so training adapts them rather than learning from scratch.

4. Handling class imbalance. With ~1.8 % positives, naïve training collapses to predicting "benign". The loss is `BCEWithLogitsLoss` weighted by `pos_weight = (1 - p)/p`, which rescales the gradient so malignant cases count proportionally — addressing imbalance at the loss level rather than by resampling.

5. Training protocol. Optimiser AdamW; mixed-precision (AMP) for speed and memory; **early stopping on validation AUC** to avoid overfitting. AUC is the primary metric because

it is threshold-independent and robust under heavy imbalance (unlike accuracy, which is meaningless here).

6. Multimodal variant (image + clinical metadata). A second model fuses the image features with a small neural network over the clinical metadata (sex, age z-scored, one-hot anatomical site, with explicit missingness flags), to test whether cheap, routinely available data improve prediction over the image alone.

7. Multi-seed robustness study (honest statistics). GPU training is non-deterministic, so a single run can mislead. Both models were retrained across **three random seeds** under an identical configuration, reporting mean ± standard deviation. This turns "model A beat model B once" into a defensible statement about the *distribution* of outcomes.

8. Operating-point selection. A classifier outputs a probability; the label depends on a threshold, which is a clinical decision, not a fixed constant. The threshold was chosen on the **validation** set via Youden's J and then applied to the **test** set — avoiding the optimistic bias of tuning the threshold on the data you report on.

9. Interpretability — Grad-CAM from scratch. Gradient-weighted class activation mapping was implemented directly (forward/backward hooks on the last convolutional layer), rather than with a library, to retain full control. It produces heatmaps of *where* the network looks when it predicts "malignant".

10. Deployment. A standalone `predict.py` runs single-image inference (CPU-only, offline) from the saved weights, with the decision threshold exposed as a parameter — showing the model as a usable component, not just a training artifact.

Results

- **Discrimination.** Image-only test **AUC ≈ 0.894** (patient-disjoint test set).
- **Operating point.** At the validation-chosen threshold (≈ 0.66): **sensitivity 0.84, specificity 0.80** — a balanced, clinically sensible trade-off. (At the naïve 0.5 threshold the model looks insensitive; the correct operating point recovers the real performance.)
- **Do clinical metadata help?** No, not beyond noise. Across seeds the multimodal model did **not** outperform the image-only baseline (ΔAUC ≈ −0.015 ± 0.025, i.e. within run-to-run variability). This negative result is reported transparently rather than hidden — on this dataset, the pixels already carry the signal.
- **Interpretability & confounders.** On confident, correct malignant cases, Grad-CAM concentrates **on the lesion** — evidence the model uses the right signal. On failures it also exposes **dataset confounders** (surgical-ink markings, rulers, hairs) that the model can latch onto — an observation that doubles as a debugging tool and a caution about ISIC-style data.

What the project demonstrates

Beyond the numbers, the project shows a way of working: leakage-controlled validation, imbalance handled at the loss, threshold selection done honestly, robustness assessed

across seeds, a negative finding reported rather than buried, and interpretability used to audit the model. The deliverables — a reproducible notebook, a packaged inference script, and a documented repository — are meant to be read and re-run, not just admired.

Stack: PyTorch, timm, albumentations, scikit-learn; executed on Google Colab. Code and figures: GitHub (github.com/SECandelo/melanoma-multimodal).